

UNIT-1

What is visualization?

Visualization is the process of representing information, data, or ideas in a visual form so that people can understand, analyze and interpret them easily.

Examples:

Chart & Graphs - bar charts, pie charts, line graphs.

Maps - showing locations or geographic data

Diagrams - flowcharts, block diagrams.

Dashboards - combining multiple visuals to show insights.

Why visualization is important:

- * Make complex data easy to understand.
- * Helps in finding patterns and trends
- * Support better decision-making
- * Improves memory and learning.

Real-life Example:

1. A weather map showing temperature with colors.
2. A student's marks graph instead of a marks list.
3. COVID-19 Dashboard showing cases visually.

History of visualization:

The history of visualization shows how humans have used pictures, symbols, maps, and graphs to understand and communicate information. It evolved from simple drawings to advanced computer-based graphics.

1. Prehistoric period (Before 3000 BC):

- * understand how visual communication evolved
- * learn the origin ~~use~~ of charts and graphs
- * Improve modern visualization design skills.

• Early humans used cave paintings and symbols. used to show hunting scenes, animals, and daily life.

Ex: Lascaux cave paintings (France), It used to communication and story-telling.

2. Ancient Civilization (3000 BC - 500 AD):

As civilizations developed, visualization become more structured & purposeful.

* Egyptians used hieroglyphs, maps and diagrams.

* Babylonians created early charts for astronomy.

* Greeks developed geometric diagrams (Euclid).

* Chinese used maps and calendars.

They are used to recording knowledge and measurements.

3. Medieval Period (500-1500):

* visualization was mainly used for religion, education, and navigation.

* Hand-drawn maps (cartography) and religious diagrams.

* Tree diagrams used for genealogy and knowledge.

* But limited accuracy due to lack of scientific tools.

4. Renaissance period (1500-1700) :

This period marked a scientific revolution. Visualization became accurate and analytical.

- * Leonardo da Vinci used detailed scientific drawings.
- * Accurate anatomical and mechanical illustrations.

5. 18th century - Birth of Modern Data visualization :

This era introduced graphs and charts for numerical data.

- * William Playfair (1786) - Invented bar charts, line graphs, and pie charts.
- line graph showing economic growth
- Bar chart comparing imports and exports
- pie chart showing market share.

6. 19th Century - Statistical visualization :

Visualization was used for social reform and decision-making.

1. Florence Nightingale used polar area charts to improve healthcare.
2. Charles Minard - visualized Napoleon's army losses during the Russian Campaign.

7. 20th century - Computer based - visualization :

~~From~~ The invention of computers transformed visualization into a digital process.

development of computer generated graphs, scientific visualization (weather models, simulations) and GIS maps for urban planning.

8. 21st Century - Interactive & Big Data visualizations

Modern visualization ~~is~~ is interactive dashboards and real-time data. It uses tools like Tableau, PowerBI, D3.js.

Ex: COVID-19 dashboards, Healthcare dashboards, business dashboards and space and climate visualization (ISRO, NASA).

Features: * Zoom, filter, drill-down.

* Real-time updates

* User Interaction.

The history of visualization shows a continuous evolution from simple drawings to advanced interactive systems, enabling humans to understand complex information and make informed decisions.

Relation between visualization and other fields

visualization is an interdisciplinary field that connects data, technology, and human understanding. It works closely with many other fields disciplines to represent information clearly and effectively.

1. Visualization and data science:

- * Data Science - focuses on collecting, cleaning and analyzing data
- * visualization is used to explore, understand, and communicate the results.

Role of visualization in data science:

- Explore Data Analysis (EDA)
- pattern, trend and outlier detection
- Explaining model results.

Ex: * using scatter plots to understand relationships between variables
* visualizing machine learning predictions using charts.

2. Visualization & UI/UX design:

- * UI (User Interface) decides how visuals are presented.
- * UX (User Experience) focuses on how users interact with visuals.

Role of UI/UX in visualization:

- layout, color schemes, typography.
- Ease of Interaction (zoom, filter, hover).
- Accessibility and clarity.

- Ex: *
- * Interactive dashboards with clear navigation.
 - * Mobile-friendly charts in apps.

3. Visualization & Computer Science:

- * provide algorithms, graphics, and rendering techniques.
- * Enables real-time and large scale visualization.

Ex: web-based charts using javascript libraries.
3D visualization and simulations.

4. Visualization & Statistics:

- * Statistics determines what should be visualized.
- * Visualization shows statistical results visually.

Ex: Histograms for data distribution

Box plots for variability and outliers.

5. Visualization & Psychology (Cognitive Science).

- * Studies how humans perceive colors, shapes, and patterns.
- * Helps avoid misleading visuals.

Ex: *

- * Using color contrast for better readability.
- * Avoiding 3D charts that distort perception.

6. Visualization & Graphic design:

- * Applies design principles to improve clarity and aesthetics.
- * Focuses on visual hierarchy and balance.

Ex: Infographics

posters and presentation slides.

visualization focuses on presenting data and information clearly, while graphic design focuses on how that information looks & feels.

Introduction to visualization

Introduction: Data visualization is the representation of data through use of common graphics, such as charts, plots, infographics and even animations. These visual display of information communicate complex data relationships and data driven insights in a way that is easy to understand.

Types of data visualizations:

Some of the common visualization techniques such as.

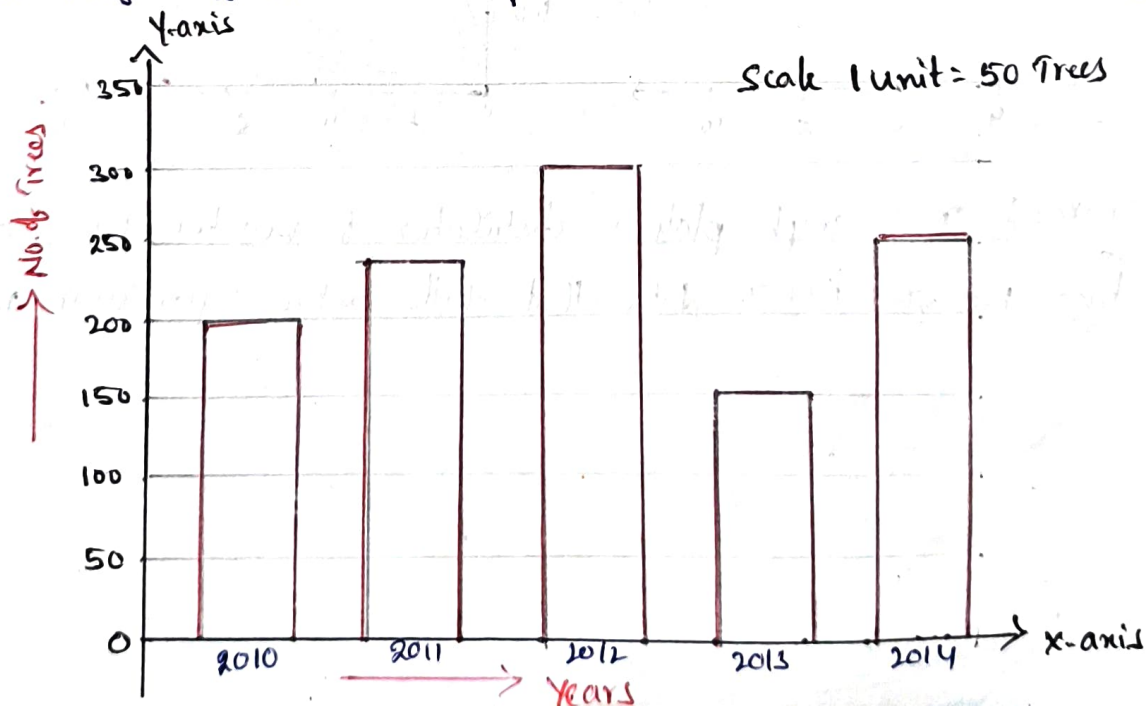
1. Tables: This consists of rows and columns used to compare variables. Tables can show great deal of information into a structured way.

Ex:

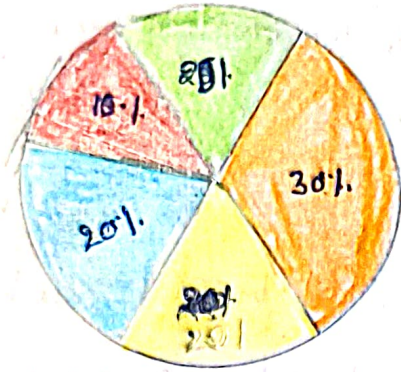
	A	B	C	D
Item-I	10	20	30	40
Item-II	5	10	15	20
Item-III	4	6	8	10
Item-IV	20	50	60	70

2. Bar charts: Bar charts are one of the most common and effective way to compare quantities across categories using rectangular bars. The length or height of each bar represents the value of the category it represents.

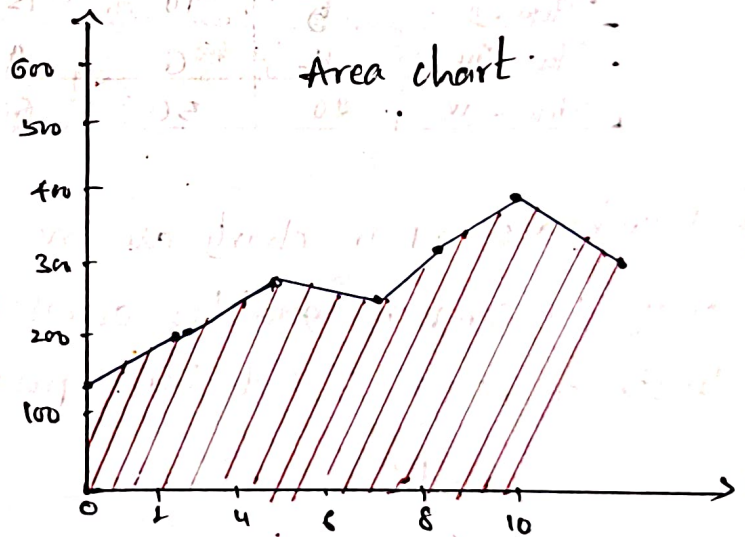
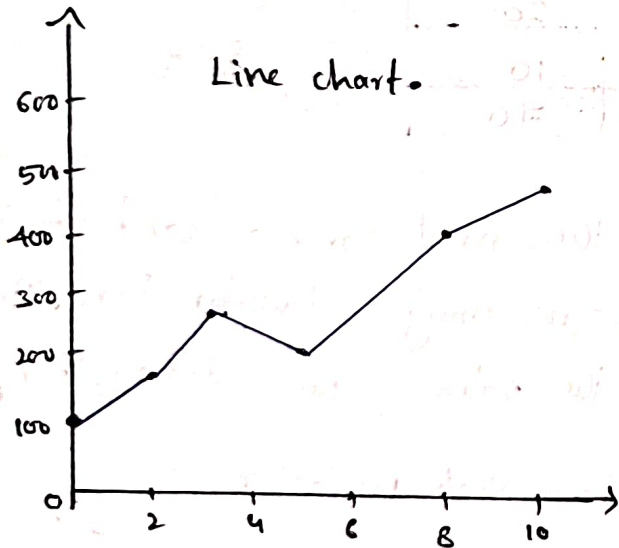
Ex:



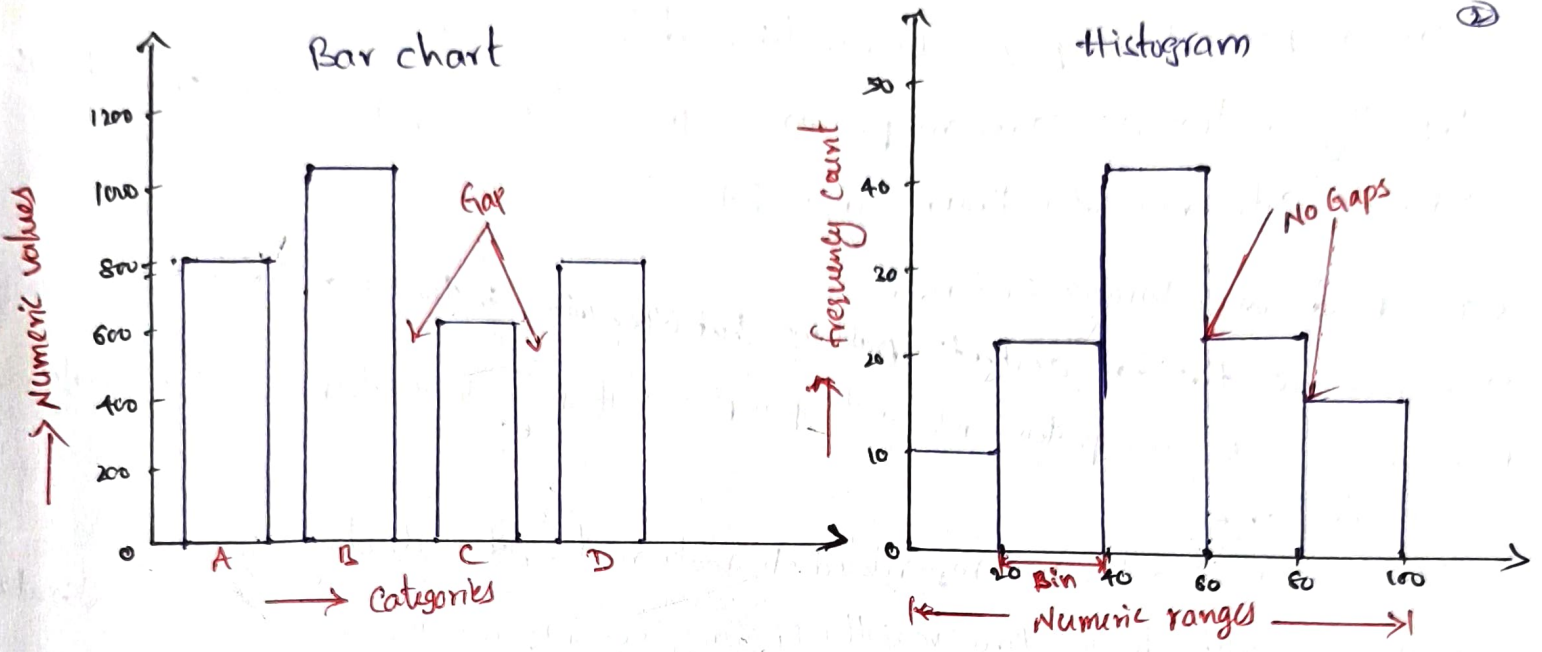
3. Pie chart and Stacked bar chart: These graphs are divided into sections that represent parts of a whole. They provide a simple way to organize data and compare the size of each component to one other.



4. Line chart and Area chart: These visuals show change in ~~over~~ one or more quantities by plotting a series of data points over time and are frequently used within the predictive analytics.



5. Histograms: The graph plots a distribution of numbers using a bar chart, representing the quantity of data that falls within a particular range.

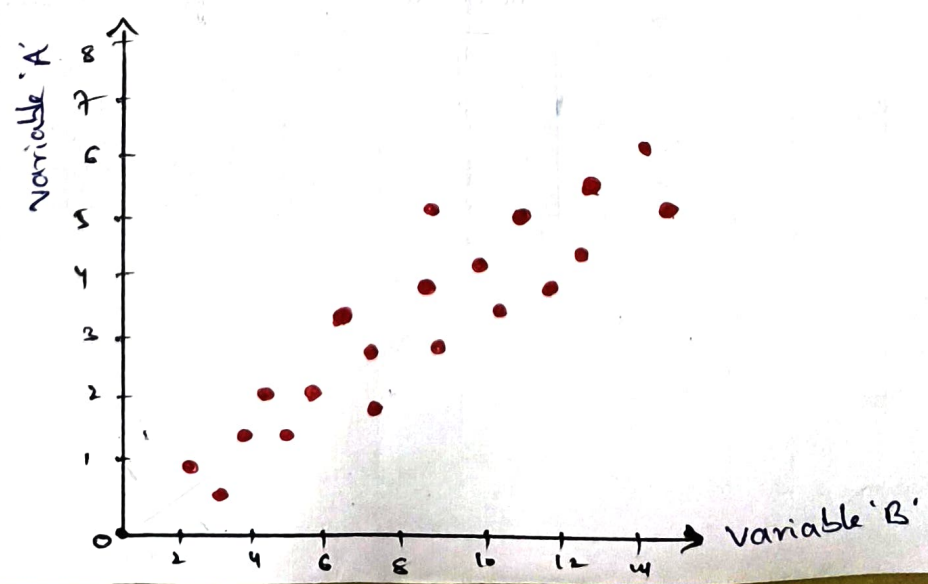


Heat maps: These graphical representation displays the magnitude of a phenomenon as color in two dimensions. They're widely used in various fields to make complex data more understandable.

Samsung Galaxy S5	8.7	8.5	9.3	9.7	8
VIVO T4x	9.2	8.3	7.3	8.7	8.7
Apple iPhone 5S	9.1	8.6	7.2	8.5	9.5
OPPO K13	8.8	9.1	9.7	9.1	8.1
	processor	Screen size	price	Battery Backup	Camera

■ Good
 ■ Average
 ■ Bad

Scatter plots: A scatter plot uses dots to represent values for two different numeric variables. The position of each dot on the horizontal and vertical axis indicates values for an individual data point. Scatter plots are used to observe relationships between variables.



8. Tree maps: It display hierarchical data as a set of nested shapes, typically rectangles. Tree maps are great for comparing the proportions between categories via their area size.

Ugly, Bad and Wrong Figures!

Ugly: A figure that has aesthetic problems but otherwise is clear and informative.

Bad: A figure that has problem related to perception: it may be unclear, confusing or overly complicated.

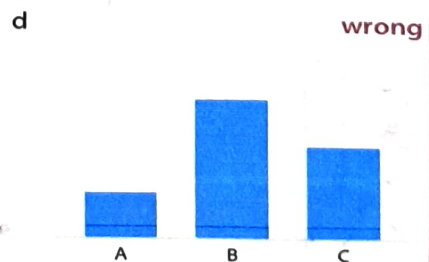
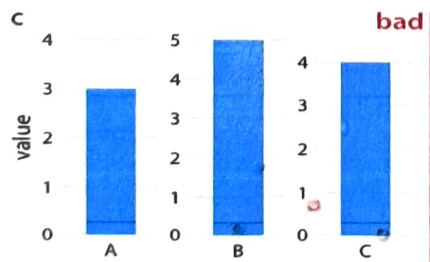
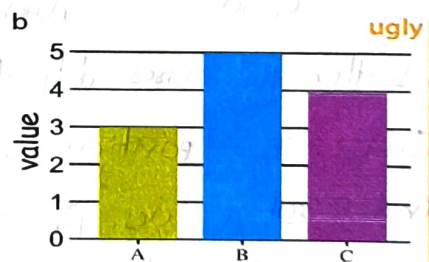
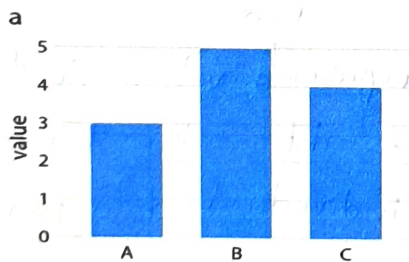
Wrong: A figure that has problems related to mathematics, it is objectively incorrect.

Figure(a): A bar plot showing three variables ($A=3$, $B=5$ & $C=4$). This is a reasonable visualization with no major flaws.

(b): An ugly version of part (a). While plot is technically correct, it is not aesthetically pleasing. The colors are too bright and not useful. The background grid is too prominent. The text is displayed using three different fonts in three different sizes.

(c): A bad version of part (a). Each bar is shown with its own y-axis scale. Because the scales don't align, this makes the figure misleading.

(d): A wrong version of part (a). Without an explicit y-axis scale, the numbers represented by the bars cannot be ascertained. The bars appear to be of lengths 1, 3, 2, even though the values displayed are meant to be 3, 5 and 4.



What is data visualization?

Data visualization is the graphical representation of information and data by using visual elements like charts, graphs and maps. Data visualization tools provide an accessible way to see and understand trends, outliers and pattern in data.

Additionally, it provides an excellent way for employees or business owners to present data to non-technical audience without confusion.

Why is data visualization important?

Modern businesses typically process large volumes of data from various data sources, such as the following.

- Internal and External websites.
- Smart devices
- Internal data collection systems
- Social media.

But raw data can be hard to comprehend and use. Hence, data scientists prepare and present data in the right context. They give it a visual form so that decision-makers can identify the relationship between data and detect hidden patterns or trends. Data visualization creates stories that advance business intelligence and support data-driven decision making and strategic planning.

What are the benefits of data visualization?

Some benefits of data visualization are as follows.

* Strategic decision-making.

Key stakeholders and top management use data visualization to interpret data meaningfully. They save time through faster data analysis and their ability to visualize the bigger picture.

② for example, they can identify patterns, discover trends, and gain insights to remain ahead of the competition.

* Improved customer service :

Data visualization highlights customer needs and wants through graphical representation. you can identify gaps in your customer service, strategically improve products (or) services, and reduce operational inefficiencies.

* Increased employee engagement :

Data visualization techniques are useful for communicating data analysis results to a large team. The entire group can visualize data together to develop common goals and plans. They can use visual analytics to measure goals and progress and improve team motivations. For example, a sales team works together to increase the height of their sales bar chart in one quarter.

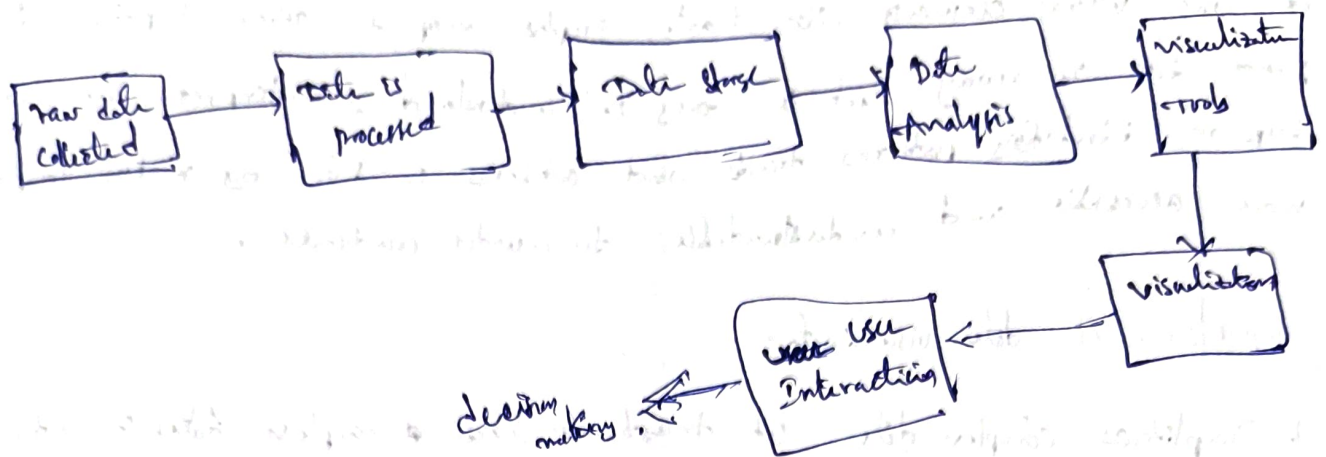
Scope of data visualization ?

Data visualization skills are highly valued in various industries, including finance, healthcare, marketing, e-commerce, government and Journalism. These industries use effective data communication to drive insights, make informed decisions and improve business performance.

(2)

(3)

Block diagram of visualization system : Stages in data visualization



1. Data collection: The initial step where data is gathered from various sources, such as data logs, APIs, sensors, or user inputs.
2. Data processing: The collected data is cleaned, transformed, and prepared for visualization. This step may involve data wrangling, normalization, and aggregation.
3. Data storage: processed data is stored in a structured format within databases, data warehouses or data lakes, making it accessible for visualization tools.
4. Data Analysis: Analytical techniques and algorithms are applied to the stored data to derive insights, identify trends, and discover patterns.
5. visualization tools: tools and software (e.g. Tableau, Power BI, matplotlib) are used to create visual representation of the analyzed data.
6. visualizations: The final visual outputs such as charts, graphs, maps, dashboards and infographics are generated.
7. User Interaction: users interact with the visualization to explore data, gain insights and make decisions.

④

* Data visualization is the graphical representation of information and data.

It uses visual elements like charts, graphs, maps & other graphical tools to present data in a way that is easy to understand & interpret. This technique helps in identifying patterns, trends and outliers in data by making complex data more accessible and understandable to a wider audience.

Importance of data visualization:

1. Simplifies complex data: It transforms large & complex datasets into visual formats that are easier to understand.
2. Identifies patterns & trends: Visualization helps in quickly spotting trends, correlations and patterns that might not be evident in raw data.
3. Enhances data interpretation: Visual representation allows for quicker comprehension, making it easier to extract insights and make data-driven decisions.
4. Improve communication: Data visualization can communicate findings effectively to non-technical audience, ensuring that key messages are conveyed clearly.

Introduction to visual perception:

visual perception refers to the way in which we interpret and make sense of visual information, such as graphs and charts. It involves understanding how different visual elements, like color, shape, and size can convey meaning and help us understand the data being presented.

Key concepts in visual perception:

Why is visual perception important?

visual perception is extremely important because it allows people to interact with the world around them. ~~while~~

What are examples of visual perception?

people use visual perception for all sorts of reasons. choosing an outfit to wear, making cup of tea, reading a book, and doing a puzzle are all activities that require visual perception.

Theories of visual perception:

There are two primary theories that seek to explain how visual perception works. They are called top-down processing and bottom-up processing. respectively. while the two theories take opposite approaches to perception, they are not mutually exclusive.

visual perception plays a critical role in data visualization. It involves understanding how humans interpret visual elements such as color, shape, size and position, which helps in designing more effective visualizations.

⑥

- Pre-Attentive processing:

This refers to the initial stage of visual perception, where the brain processes certain elements of the visual field almost instantaneously without conscious thought. Features like color, shape, size and orientation are detected pre-attentively.

- Gestalt principles:

These are the set of principles that describe how humans naturally group similar elements, recognize patterns, and simplify complex images.

- Color perception:

Color can influence how we interpret data. Some color combinations are easier to distinguish than others, and color can convey specific meanings (e.g. red for warning, green for go). Understanding color theory and colorblind accessibility is important for effective visualizations.

- Visual hierarchy:

This is the arrangement of elements in a way that implies importance. By using different sizes, colors, and placements, designers can guide viewers through the information in the intended order.

Applying visual perception principles to visualization:

- Use of pre-Attentive Attributes:

Employ visual features like color, size and shape to highlight important data points or trends. For example using a distinct color to highlight key data points in a scatter plot can draw immediate attention to those points.

- Leveraging Gestalt Principles:

✓ Design charts and graphs that naturally guide the viewers' perception.

Explain preattentive processing with an Example.

Pre-attentive processing refers to the brain's ability to quickly and automatically detect certain visual features without conscious effort. This helps in rapidly identifying and responding to important elements in a visual scene.

Example: Detecting an outlier in a scatter plot where most of the data points are blue circles, representing various data values. Among these blue circles, there is a single red circle.

Pre-attentive Features in Action:

1. Color Difference: The red circle immediately stands out against the blue circles.
2. Automatic Detection: You can instantly notice the red circle without having to consciously search for it.
3. Speed: This recognition happens almost instantaneously, within a fraction of a second.

Why it works:

- * color: Human vision is highly sensitive to color differences. A unique color among a uniform set captures attention immediately.
- * contrast: The red circle contrasts sharply with the blue circles, making it pop out visually.

Application in data visualization:

By utilizing pre-attentive processing principles, designers can highlight important information effectively. For instance, in the scatter plot example, the red circle could represent an outlier or a critical data point that needs attention. By making it visually distinct, viewers can quickly identify and focus on this important element without extra effort.

Example: Identifying a trend in a line chart:

Imagine a line chart showing sales data overtime, where the overall trend is represented by a continuous blue line. To highlight a significant increase in sales during a particular period, you might use a different color, such as green, for that segment of the line.

Pre-attentive features in Action:

1. color change : The change from blue to green immediately draws attention to that specific segment.
2. Orientation : If the slope of the line increases sharply, the change in orientation also aids quick detection.

In this way, pre-attentive processing helps viewers quickly grasp the essential parts of the data, improving the efficiency and effectiveness of visual communication.

- proximity: Group related data points closely together.
- Similarity: Use consistent colors and shapes for similar data categories to reinforce grouping.
- closure & continuity: Ensure lines and shapes are complete and continuous to make patterns and trends more apparent.

• Effective color use:

Choose color palettes that are both aesthetically pleasing and functionally effective. Avoid using too many colors, which can confuse the viewer, and ensure that colorblind-friendly palettes are considered.

• Creating visual hierarchy:

Use visual hierarchy to optimize information. Larger, bolder, or brighter elements will draw attention first, so use these to highlight the most important datapoints or messages.

• Simplifying Complex Data:

Break down complex data into simpler visual elements. Use white space effectively to avoid clutter and ensure that the visualization is easy to navigate.

Examples of visual perception in action:

1. Heat maps: By using color gradients, heat maps leverage pre-attentive processing to allow quick identification of areas with higher or lower values.
2. Line Graphs: Continuity is used to show trends over time, making it easier to follow the progression of data.

③

3. Scatter plots: Similarity and proximity help in identifying clusters or correlations between variables.

Visual representation of data:

- The visual representation of data involves transforming raw data into graphical formats that facilitate understanding, analysis and communication.
- Visual representation of data involves transforming raw data into visual formats such as charts, graphs and maps. The goal is to make data comprehensible and actionable by ~~highly~~ highlighting patterns, trends and outliers that might be missed in textual or numerical representations.

Data Abstraction:

• Data abstraction is the process of simplifying complex data into more manageable and understandable forms without losing essential information. This can be achieved through techniques such as:

1. Aggregation: Summarizing data to reduce its volume (eg. averages, totals).

2. Filtering: Selecting a subset of data that is most relevant to the analysis.

3. Transformation: Changing the data format or structure to make it more suitable for analysis (eg. normalizing values, calculating ratios).

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Visual perception: Visual perception in the context of data visualization refers to the process by which the human brain interprets and understands visual elements presented in a visual format. It involves how viewers recognize patterns, detect anomalies, and extract meaning from visual elements like shapes, color, sizes, and spatial arrangements. In data visualization, visual perception determines how easily and accurately users understand data from charts & graphs.

Importance in data visualization:

1. Faster understanding:

Visual perception enables users to interpret large and complex data quickly without reading through raw numbers.

Scenario: An electricity board provides monthly energy consumption data for 12 months.

→ Raw Table (Hard to understand quickly).

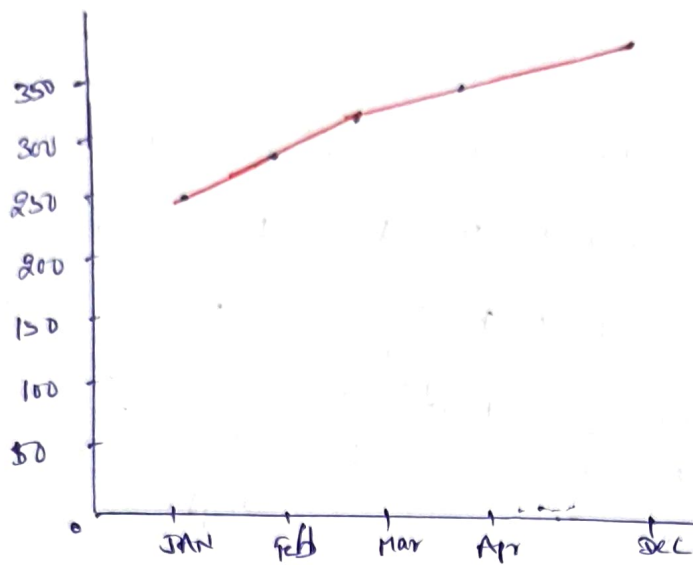
Month	Unit Consumed
Jan	250
Feb	280
Mar	300
...	
Dec	350

You'd have to read all 12 numbers, compare them mentally, and then figure out trends.

→ Line chart (visual perception makes it fast)

* A line chart shows a rising trend from January to December.

* The eye catches the slope without reading exact value.



Q. Electricity usage is steadily increasing over the year.
pattern detection:

Helps in identifying trends, clusters, and outliers easily using visual cues like color and shape.

Scenario: A company tracks sales performance of 50 employees across different regions.

→ without visualization (Hard to detect patterns)

- * You have a spreadsheet of names, sales numbers, and regions.
- * It's quickly difficult to quickly tell who is performing well or where clusters of high performance exist.

→ with visualization - Example: bubble chart.

- * Each bubble represents one employee.
- * x-axis: Number of deals closed.
- * y-axis: Total sales amount.
- * color: Represents region.
- * size: Represents revenue generated.

Visual perception at work:

* trend: you^{ll} notice that larger bubbles generally lie along a rising diagonal

"Higher deals = Higher sales

* clusters: Employees from Region A (Blue) form a tight group.

* outliers: one red bubble far from others → Top performer or data error.

Insight: ~~you~~ we can see patterns instantly without scanning through all 50 rows of data. — because color, position and size guide your perception automatically.

3. Reduces cognitive load:

A well-designed visual leverages perception to minimize mental effort and improve clarity.

Scenario: A project manager wants to track the status of 10 ongoing projects, each with deadlines, progress, and priority levels.

project	deadline	progress	priority
Alpha	15 July	75%	high
Beta	20 July	40%	medium
Gamma	18 July	90%	high

* The user has to read every row, compare percentages, and interpret deadlines mentally.

* This increases cognitive load, making it slower and more error-prone.

② with visualization (progress Dashboard - low cognitive load)

- * progress Bars to show % completion.

- * color coding:

Red = High priority

Yellow = Medium

Green = Low

- * Icons or flags to highlight overdue or urgent projects.

Visual perception in Action:

- * viewers can instantly see which projects are nearly complete, which ones are delayed, and which are urgent.

- * No need to read all rows - the color and layout guide the viewers attention.

Insight: A clean, visual layout reduces the mental effort needed to interpret complex data.

4. Effective Communication:

By aligning design with visual perception principles, visualizations can convey insights more intuitively and persuasively.

Scenario: we are presenting election results for four political parties across five states.

1 → Raw Data Table (Hard to communicate at a glance):

State	party A	party B	party C	party D
TN	40	30	25	10
KA	40	35	20	5
AP	50	20	20	10
KL	35	40	15	10
MH	48	25	15	12

Even though the data is there, the insights are not obvious unless someone reads every number carefully.

2 → stacked bar chart with color-coding (Highly effective visual).

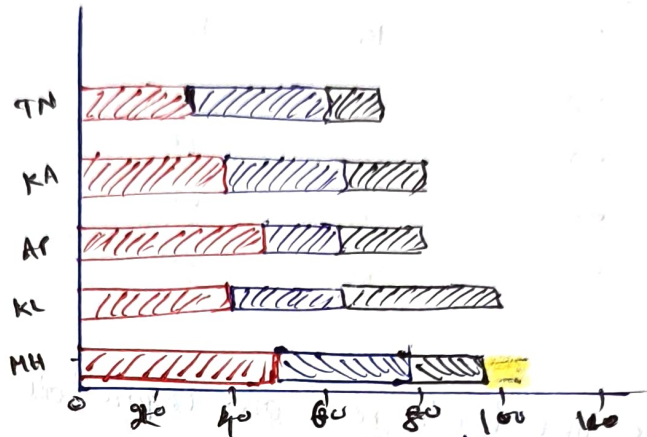
Each state is represented by a stacked horizontal bar, and each party is color-coded.

Red - party A

Blue - party B

Green - party C

Yellow - party D



* Instantly see that party A leads in most states

* The proportions are clear because color + positions are easy to process visually.

* we don't need to read numbers to understand the result.

Visual perception principles at work:

* color helps differentiate parties

* size/length of segments shows dominance

* layout makes comparison across states easy.

Insight delivered clearly:

"party A is leading in all five states, especially strong in Andhrapradesh and Maharashtra."

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Reduce

3 Cognitive loading example:-

Scenario:- You want to display the weekly weather forecast including temperature, rain chance & wind speed.

1. Data table

Day	Temp (°C)	Rain %	wind (km/h)
Mon	36	10	18
Tue	37	80	22
Wed	34	30	15
Thu	38	5	25
Fri	33	60	20

The user has to read and compare all numbers - this increases mental effort

2. Icon-based weather chart (reduce cognitive load)

* Each day is shown with weather icons ☀️ ☁️ ➡️

* Temperature is shown using a color bar:

→ Red for hot day

→ Blue for cooler day

* Rain probability shown using droplet icons (000)

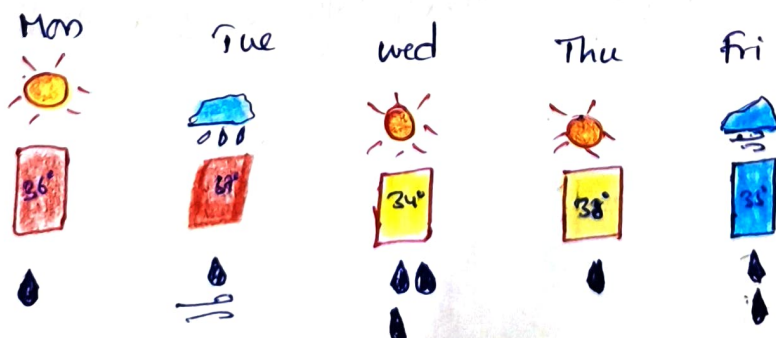
* ~~Users~~

users don't need to read every number - they instantly see.

→ which day will be hottest (deep red)

→ which day has the most rain (more 0 icons)

→ windy days (➡️ symbol).



Role of color in Data visualization:

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Color plays a crucial role in visual representation, significantly impacting how information is perceived and interpreted.

1. Emphasizing Important Data:

Eg: Using a bright or contrasting color to highlight critical data points, such as a spike in sales or a significant drop in performance, helps draw immediate attention to these areas.

2. Improving Readability & Clarity:

Eg: High-contrast color for text and background. Improve readability for instance, using dark text on a light background or vice-versa ensure that labels and numbers are easily readable.

3. Categorizing and Grouping Data:

Eg: Different colors can be used to distinguish between data categories in a bar chart or segments in a pie chart. For example, using distinct colors for different departments in a company's performance chart helps quickly identify each department's contributions.

4. Conveying Trends & Patterns:

Eg: Gradient colors can show trends and patterns over time or across categories. For instance, a heatmap might use a gradient from blue (low values) to red (high values) to depict temperature changes across different regions.

5. Highlighting Comparisons:

Eg: Using contrasting colors to compare data sets makes it easier to spot differences and similarities. For example, a side-by-side comparison of two products' sales figures using different colors helps visually separate and compare the data.

6. Guiding user focus:

Eg: color can guide the viewer's focus towards specific parts of the visualization. using a muted color palette with a single accent color can highlight key data points without overwhelming the viewer.

7. Enhancing Aesthetic Appeal:

Eg: A well-chosen color palette can make a visualization more aesthetically pleasing, encouraging engagement and making the data more enjoyable to explore. For example, using a harmonious color scheme based on color theory can create an attractive and professional-looking chart.

8. Aiding memory Retention:

Eg: consistent use of specific colors for certain data types or categories across multiple visualizations helps user remember and quickly recognize these categories. For instance, always using green for profits and red for losses across financial reports aids quick recognition.

9. Avoiding Misinterpretation:

Eg: poor color choices can lead to confusion. Using colors that are too similar can make it hard to distinguish between data points. For example, using two shades of blue that are very close in tone might confuse viewers.

10. Ensuring Accessibility:

Eg: using colorblind-friendly palettes and combining colors with patterns or labels ensure that everyone, including those with color vision deficiencies, can interpret the data accurately.

By carefully selecting colors, you can enhance the clarity, effectiveness and accessibility of data visualizations, making complex information more understandable and engaging for a wide audience.

Gestalt principles of visual perception:

The Gestalt principles explain how humans naturally perceive patterns and organize visual elements into groups or unified whole. These principles are widely applied in data visualization to design more effective and intuitive graphics.

1. proximity
 2. similarity
 3. Enclosure
 4. closure
 5. continuity
 6. Figure-ground
- Common facts

1. proximity: proximity refers to the tendency of our brain to group elements together when they are placed close to each other in space.

In data visualization, designers use proximity to show relationships between data points or groups. Objects placed near each other are seen as related, while more distant objects are perceived as unrelated.

Example: In a scatter plot, if the data points are clustered tightly in one area, viewers naturally perceive those points as belonging to the same group or category.

without proximity:

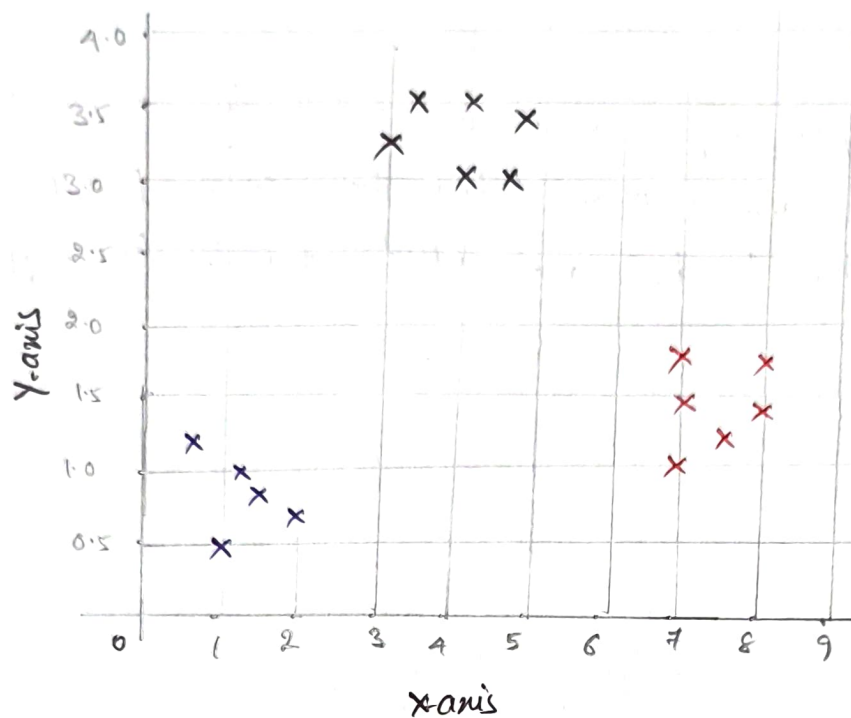


(No clear grouping)

with proximity:



(Three distinct groups are perceived.)



x - cluster-1

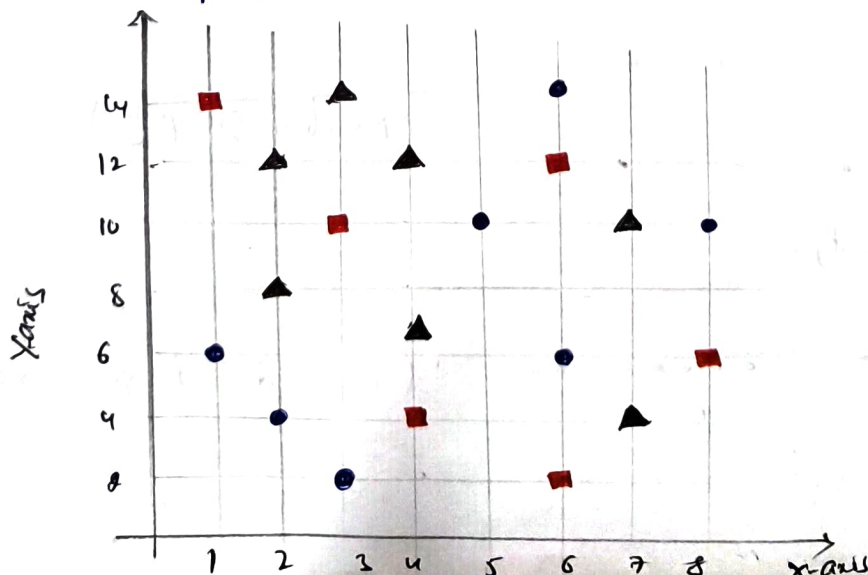
x - cluster-2

x - cluster-3

2. Similarity: The principle of similarity states that elements that share similar visual characteristics. In simple words, it means that when we see similar things we usually group them together. Such as color, shape, size, or texture. etc. group them together.

In data visualization, designers use similarity to help viewers quickly identify and group related data points based on shared visual features.

Example: In a scatter plot, when plotting multiple ~~cat~~ categories of data, using distinct color or shapes for each category allows users to easily distinguish and group the data points.



● - Category A (blue circles)

■ - Category B (red squares)

▲ - Category C (black triangles)

(20)

3. Enclosure (Common Region): This principle of Enclosure, also called common region, states that objects placed inside the same boundary or region are perceived as belonging together.

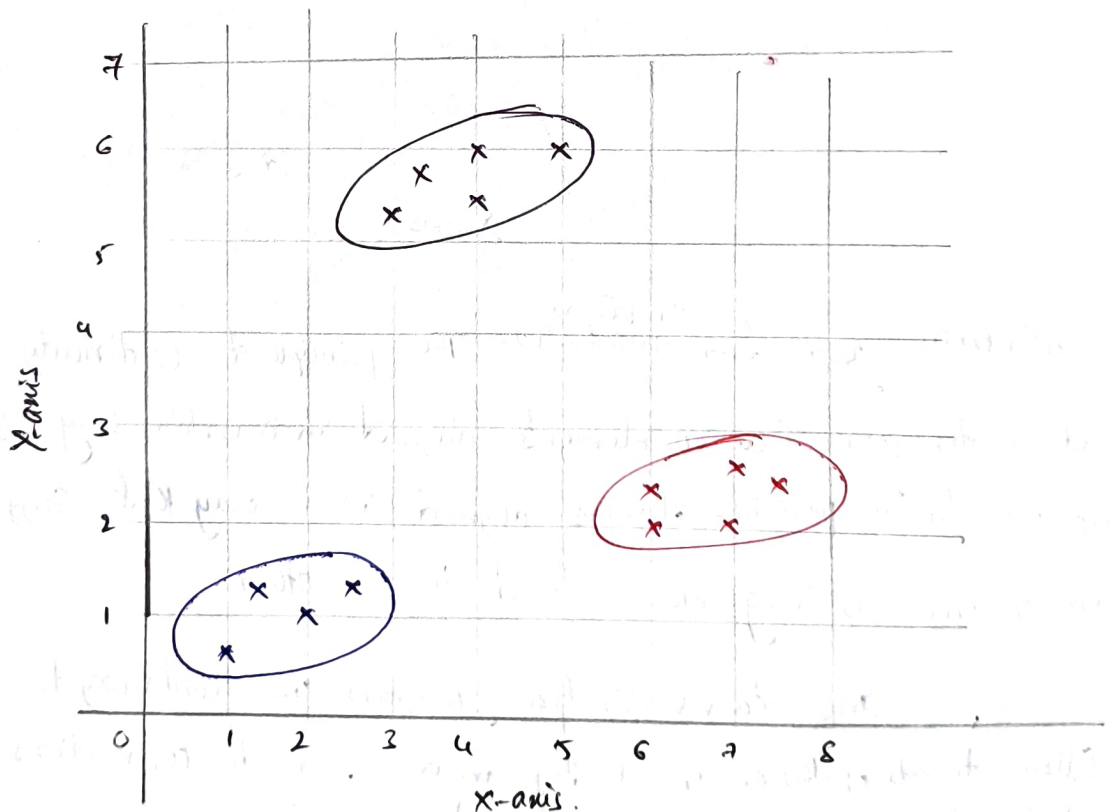
In data visualization we use boxes, borders or shaded regions to group related data elements and distinguish them from unrelated ones. This makes complex visualization more organized and easy to interpret.

Example: on a dash board, there may be multiple charts

- * A light gray box surrounding KPIs (key performance indicators) like revenue, cost, and profit visually group them as "financial metrics".

- * Another box encloses charts for customer demographics, indicating a separate group.

Real world Example: In a scatter plot, drawing an ellipse or rectangle around a cluster of points emphasizes they are part of a specific group or category (e.g. high priority customers).

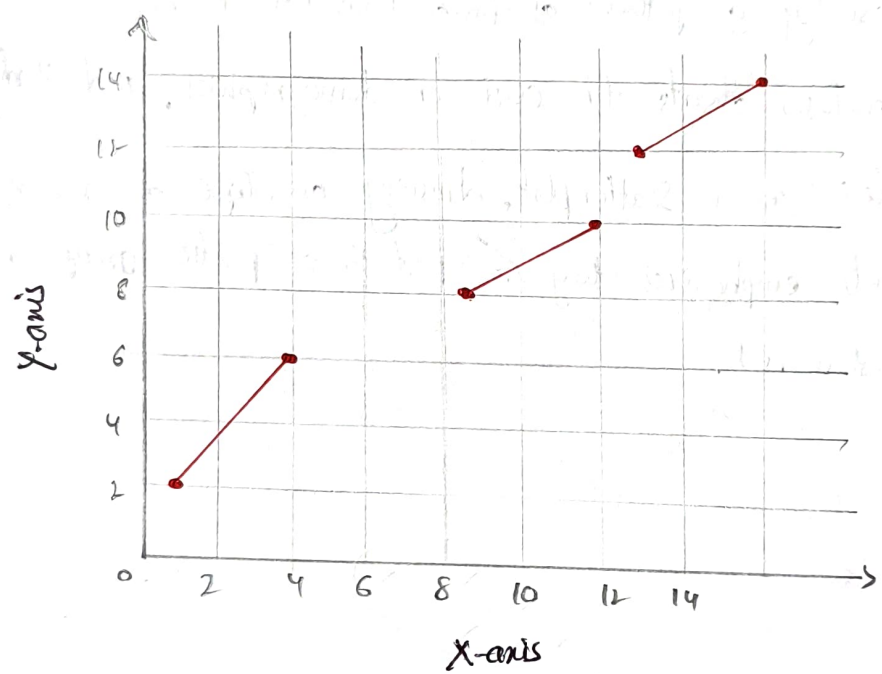


4. closure: The principle of ~~close~~ closure states that when parts of a visual element are missing or incomplete, the human mind tends to fill in the gaps and perceive it as a ~~for~~ complete, unified shape.

In data visualization, designers use closure to simplify visuals and help viewers focus on key parts of the data without drawing all the details explicitly.

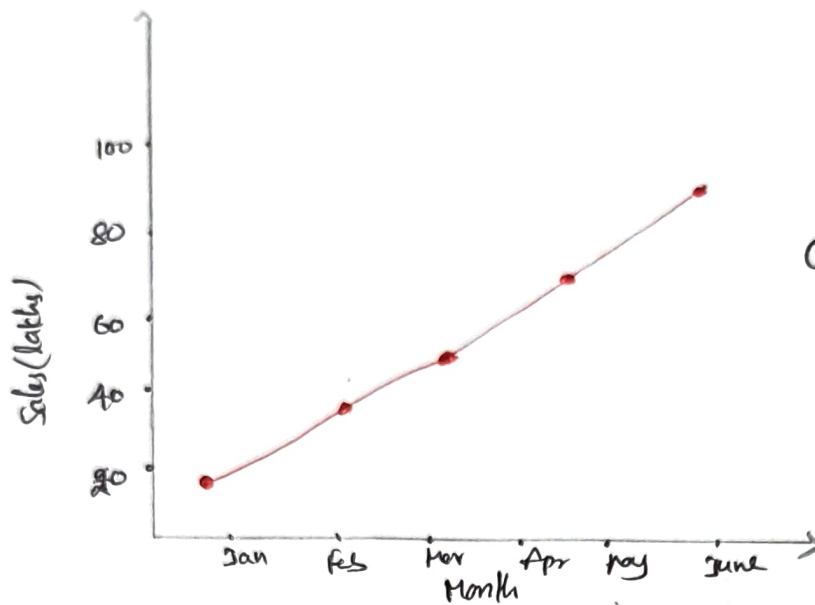
Example: * In a line chart, if parts of the trend line are missing due to incomplete data, the viewers mind fills in the missing section and still perceives the general trend.

* If a pie chart has gaps between slices, we perceive it as a circle.

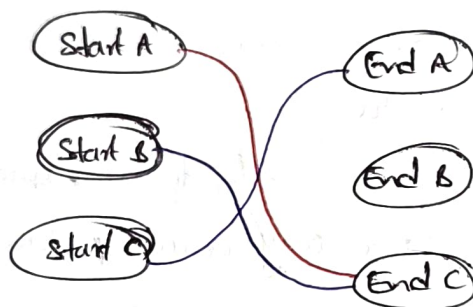


5. Continuity (Good Continuation ~~Communication~~). The principle of Continuity, also called Good Continuation, focus on elements aligned in a certain way. It states that the brain perceives elements aligned in a way that suggest a continuous line or curve as being more related to each other.

In data visualization, designers use continuity to help viewers easily follow trends, paths or relationships in data. Smooth connections make it more intuitive for users to interpret information.



Continuity in line chart



Continuity in Flow Diagram

G. Figure - Ground: The figure ground principle states that the human brain separates visual elements into two parts:

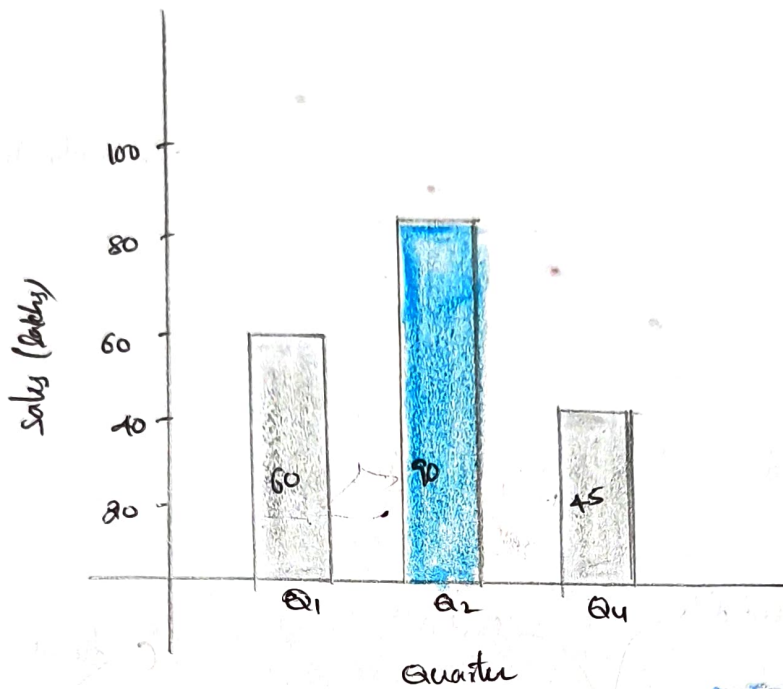
* Figure: The object of attention, perceived as being in the foreground.

* Ground: The rest of the visual field, perceived as behind the figure.

This distinction helps viewers focus on the most important elements and ignore irrelevant background details.

In data visualization, designers use figure ground contrast to highlight key data and de-emphasize less important elements. By manipulating color, brightness, and boundaries, they guide the viewer's focus.

Example: In a bar chart, one bar is shown in a bold, saturated color (Figure) while the other bars are in light gray (Ground). This makes the highlighted bar immediately stand out.

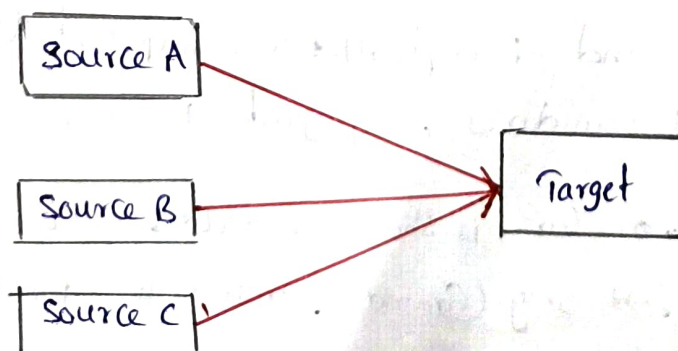


7 Common fate: The common fate principle states that elements moving in the same direction or changing in the same way are perceived as part of the same group.

In data visualization, designers use common fate by aligning visual elements along a shared path or direction to indicate relationships or grouping. This can also involve elements that change together over time.

Example: In an animated line chart - two lines moving upward together (in the same slope and direction) suggest the variables are related.

(or) in a flow diagram - multiple arrows pointing towards the same target indicate processes leading to a common outcome.



Why is Visual Encoding important in data visualization: (25)

visual encoding is crucial in data visualization because it directly impacts how effectively information is communicated and understood. Here are some key reasons why visual encoding is important.

1. Clarity and Comprehension:

- * proper visual encoding makes complex data more accessible and understandable.

- * It highlights patterns, trends, and anomalies, making it easier for users to interpret the data quickly. (e.g. Tallest bar in a chart instantly shows the highest sales)

2. Efficiency:

- * visual encoding allows viewers to process large amounts of data swiftly.

- * It leverages human perceptual abilities, enabling immediate recognition of visual patterns.

3. Accuracy:

- * Effective visual encoding ensures that data is represented truthfully without distortion.

- * It helps avoid misleading interpretations by using appropriate visual elements. (E.g. Using area (circle size) instead of bar length for comparing close values can mislead, only aligned bars for comparing product revenues.)

4. Engagement:

- * well encoded visualizations are more engaging and can capture the audience's attention.

- * They encourage users to explore and interact with the data leading to deeper insights.

(15)

5. Comparison:

- * visual encoding facilitates easy comparison between different data points or categories.
- * It allows user to make relative assessment and identify differences and similarities. (eg: sorting bars from highest to lowest makes comparison easier).

6. Communication:

- * clear visual encoding enhances communication by conveying information. Succinctly.
- * It aids in storytelling, helping to present a compelling narrative based on data.

7. Decision making:

- * By providing a clear view of the data, visual encoding supports informed decision-making.
 - * It helps stakeholders identify key factors and make data driven choices.
- (eg: A dashboard using position and color to flag underperforming departments).

Example for visual Encoding - Sales data visualization.

Let's say we want show sales for four product categories:

Category	Sales (\$M)
Electronics	150
Fashion	100
Grocery	80
Furniture	120

we can use:

- * Bar length: to encode sales amount
- * color hue: to differentiate product categories
- * position: to align bars along a common axis for easy comparison

(27)

• Data abstraction is the process of reducing complexity by highlighting essential features while suppressing irrelevant details. This involves summarizing and simplifying data to focus on the key aspects that need to be communicated. Levels of data abstraction include:

1. Raw data: The unprocessed, detailed data.
2. Data Tables: Organized collections of data in rows and columns.
3. Aggregated Data: Summarized data, such as averages or totals.
4. Visual Representation: Graphical displays like charts, graphs, and maps that provide a visual abstraction of the data.

Visual Encodings:

Visual encoding is the method by which data attributes are mapped to visual elements. Effective visual encodings help viewers quickly and accurately interpret the data.

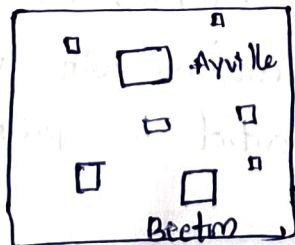
Visual encodings refer to the use of visual elements to represent data values. Key elements include:

1. Position: The spatial placement of elements (e.g. points on a scatter plot).

Generally, graphs map across the x and y-axis. They work great to present any quantitative data. It deals with the flat screens and just two planar variables.

2. Size: The magnitude of elements.

In general, the size indicates greater quantity or importance.

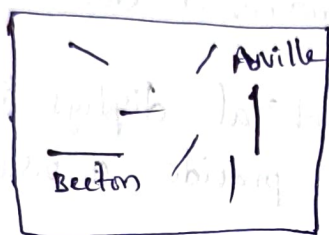


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Eg. figure shows Ayville has a greater population than Beeton.
The bigger the square size, the longer is the quantity.

3. orientation : the direction of elements;

orientation typically indicates relative orientation, direction of flow or movement. From the given example, it is clearly seen that Ayville is perpendicular to Beeton.



4. Shape : The form of elements (e.g. different shapes for different categories)..

Shapes may be used simply to distinguish between different symbols. The symbols can be either abstract symbol or mimetic symbol.

The mimetic symbol is also termed to be iconical. An icon is a small picture that provides a metaphor for particular idea or process.



Abstract symbol



Mimetic symbol.

5. Color : color is often modeled as three components - HSV model.

Hue : indicates the redness or greenness of the color that it represents. Many hues have particular association.

Saturation : or chroma, describes about the color purity. Saturation is often used in combination with values, but may also be used independently to control the prominence of symbols.

value gives the color intensity as lightness or darkness. (29)
value is often used to like size to indicate quantity or importance.

6. Texture : patterns used to differentiate areas or groups.

7. Marks : Marks is a basic graphical element or a geometric primitive. It can be represented as point, line (1D), area (2D), volume (3D).

Use of color :

color is a powerful tool in data visualization but must be used thoughtfully to enhance understanding without causing confusion.

1. Differentiate categories : use distinct colors to represent different groups or categories.

2. Indicate values : Apply color gradients to show data ranges or intensities.

3. Highlight Important Data : use contrasting colors to draw attention to key points or anomalies.

* perceptual issues:-

understanding perceptual issues is essential for effective visualization.

Common issues include:

1. color perception : variation in how people perceive colors can affect the readability of visualizations.

2. visual clutter : overloading a visualization with too many elements can make it difficult to interpret.

3. Misleading Representations : Inaccurate visual representations, such as distorted axes or inappropriate scales, can lead to incorrect interpretations.

4. Context Loss: Simplifying data too much can result in the loss of important context, making the visualization less informative.

Gestalt principles:

Gestalt principles describe how humans group similar elements. They also help explain how we recognize patterns and understand complex images.

1. Figure-Ground

2. Similarity

3. Proximity (or) Common Region

4. Continuity

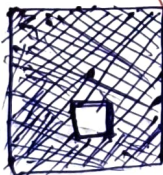
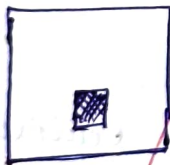
5. Closure

6. Symmetry and Order

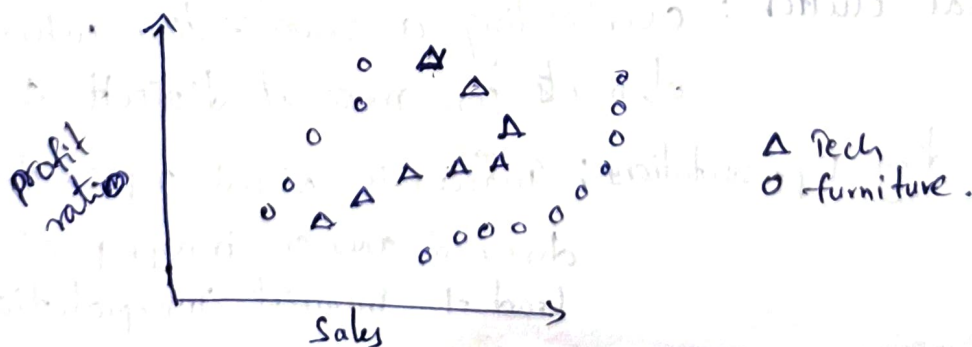
Information overload in data visualization occurs when a visual display presents too much data, too many visual elements, or excessive detail, making it difficult for us to understand the message, or extract insight.

~~no need~~

1. Figure-Ground: This rule states that people when they see a figure either see the foreground or background.



2. Similarity: In simple word it means that when we see similar things we usually group them together. Same color, shape, etc. we group them together.



Information overload in data visualization:

(30)

Information overload in data visualization occurs when the presentation of data includes more information than the viewer can effectively process and understand. This can lead to confusion, misinterpretation, or outright ignoring the data.

Contributing Factors (Causes of Information overload in data visualization)

1. **Excessive data points:** including too many data points can overwhelm viewers. (E.g. line chart with hundred of values on the x-axis)
2. **Complex Graphics:** overly complex charts and graphs can difficult to interpret. (E.g. multiple charts squeezed into a small dashboard).
3. **Irrelevant Information:** including non-essential data can distract from the key message. (E.g. including unnecessary labels, legends or tables).
4. **Poor design:** Inconsistent colors, fonts, and layout can make visualization harder to read. (E.g. using multiple bright colors, patterns and fonts in a single chart).
5. **Lack of focus:** Not ~~for~~ highlighting the most important information can cause viewers to miss the main points.

Effects of Information overload

Mitigation strategies:

1. **Misinterpretation of data**
 2. **Increased cognitive load**
 3. **Slower decision making**
 4. **User frustration**
1. **Simplify visuals:** use simpler charts and avoid unnecessary decoration (chartjunk).
 2. **Highlight key data:** use color, size or annotations to draw attention to the most important ~~data~~ information.
 3. **Limit data points:** show only the most relevant data points and consider using interactive elements to allow viewers to explore additional details if needed.

③

4. use clear labels: Ensure all labels, legends, and titles are clear and concise.
5. consistent design: use a consistent color scheme, fonts, and layout to improve readability.
6. storytelling: structure the visualization to tell a clear and compelling story, guiding viewers through the data logically.

Example: A Company Sales Dashboard:

1) Poor Design (overloaded): A company's quarterly sales dashboard displays:

- * 15+ charts on a single page
- * pie charts with over 12 slices each
- * Bright, clashing colors
- * Repetitive data in both tables and charts
- * Tiny fonts and dense labels.

Result: Managers struggle to find insights quickly. It takes extra time to interpret the data, increasing decision-making delay and errors.

2) Good design:

- * 4 key charts: sales trend, top products, region-wise comparison, and Target vs Actual.
- * clear use of color (one color palette with highlight for underperformance)
- * tooltips instead of cluttered labels.
- * summary KPIs shown at the top.

Result: Stakeholders immediately grasp the trends and can take informed actions quickly.